# Lab 16 - Post-Processing - Support Removal and Sanding

Course: Innovating Product Development with 3D Printing and Additive Manufacturing (TGS-2022017597) Topic 03: Process Improvement for AM TSC objective: A5 - apply post-processing to lift the quality of an FDM part

## Goal

Print the vase tilted so it needs supports, remove them cleanly, then sand the scarred surfaces through progressive grits to a smooth finish.

## What you'll build

A tilted-print vase taken from raw support-scarred print to a smooth finished part.

Tools: Ultimaker Cura, open vase.stl, test-part.stl, FDM printer, pliers, sandpaper 120→400 grit

## Steps

1. Open 'open vase.stl' in Cura and rotate it 15-20° off vertical so one side overhangs.

```
Rotate - 15-20 degrees  tilt
```

2. Enable supports (Touching Buildplate) and slice.

```
Support > Generate ✔ > Touching Buildplate
```

3. Print the tilted vase with its supports.

```
Save to Disk > print
```

4. Snap the supports off with pliers, working from the outer edge toward the part.

```
Needle-nose pliers
```

5. Sand the support-scarred areas with 120-grit, then step through 240 -> 400 grit in circular motions.

```
Sandpaper 120 -> 240 -> 400
```

6. Optionally wet-sand at 400 grit for a satin finish, and compare the treated side with an untouched area.

```
Wet sand + compare
```

7. Inspect test-part.stl in Cura and write a finishing plan before printing: support access, tool order, surfaces to protect and dimensions that sanding must not change.

```
Open test-part.stl > Preview supports > annotate finishing plan
```

## Evidence to capture

- Annotated before/after photographs of support scars and the sanded surface.
- Finishing log with tools, grit sequence, time and observed dimensional/detail loss.

## Test it

The support side is smooth from a normal viewing distance, and your evidence shows a planned 120->240->400 grit process that removes scars without rounding protected features or critical dimensions.

## Files in this folder

- open vase.stl - Core sample model used by the lab procedure.
- test-part.stl - Mixed-feature benchmark for planning support removal and progressive finishing.